Earth-observation link report

SENTINEL-2C -> Glasgow (Strathclyde)

Generated 2026-11-06 20:29:23 UTC. Simulation clock 2026-11-06 20:29:23 UTC. Payload: Frog fibres, one frame (EO sim) (159 bit). Station network: all (39 sites with X-band, elevation mask 5 deg). No target: the message is created on board now.



Globe at the simulation time: selected satellite (yellow diamond) with its footprint at the elevation mask and ground track; stations (cyan, white when in view); data centre (magenta) with the fibre arc from the receiving station; target (orange); GEO relays (yellow, far out).

1. The satellite

item	value
Name / NORAD / COSPAR	SENTINEL-2C / 60989 / 2024-157A
Family / operator	Sentinel-2 / ESA / Copernicus
Sensor / resolution / swath	MSI, 13 bands 443–2190 nm; 10 / 20 / 60 m; 290 km
Position at simulation time	lat 55.877 deg, lon -137.056 deg, alt 797.7 km, 7.460 km/s
Orbit	incl 98.567 deg, period 100.58 min (14.317 rev/day), perigee 785 km, apogee 787 km, e 0.000087
Elements	epoch 2026-11-05 15:23:04 UTC (age 29.1 h); CelesTrak GP data, SGP4 propagation
TLE line 1	1 60989U 24157A 26309.64101987 .00000056 00000+0 38017-4 0 9990
TLE line 2	2 60989 98.5673 22.3880 0000873 99.2986 260.8296 14.30824878113212

Payload downlink (SOURCED)

parameter	value
Band / centre frequency	X-band, 8.0900 GHz
Nominal data rate	520.0 Mbit/s
Modulation / coding	8PSK; 3 bit/symbol, code rate 0.667; required Eb/N0 6.5 dB
Satellite EIRP	24 dBW (ASSUMED - not published)
TT&C channel	S-band 2250 MHz, 2.0 Mbit/s
Direct broadcast	none
GEO relay	EDRS, 1.80 Gbit/s - Laser terminal to EDRS; near-real-time delivery to Harwell / Weilheim / Redu gateways
Sources	ESA Sentinel-2 mission: 520 Mbit/s X-band downlink, EDRS 1.8 Gbit/s | ESA EDRS

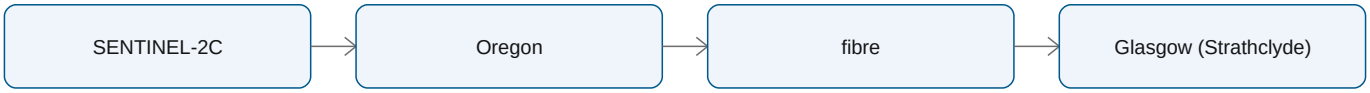
2. The communication path and where the time goes

Message creation time t0 = 2026-11-06 20:29:23 UTC: message created on board now. Payload 159 bit (eo_satellite_sim SUMMARY.md: 159 bit/frame). Every route below is computed for this payload from the 48 h contact prediction; a payload larger than one pass can carry is drained over successive passes.

Recorder queue estimated from the ESA Sentinel-2 acquisition plan (live file s2c_mp_acq__kml_20261029t120000_20261116t150000): 0.0 GB now; 1015 GB acquired in the last 24 h at 490 Mbit/s compressed; peak in the last 24 h 51 GB; recorder next empty at 21:05 UTC via EDRS-C; last datatake 19:27 UTC, 2.3 min, 39 scenes; next 20:46 UTC. Service model: 560 Mbit/s X-band over Matera, Maspalomas, Svalbard, Inuvik; EDRS 1.8 Gbit/s up to 25 min per orbit while in line of sight; 2.4 Tbit memory.

On-board queue: 0.0 GB already in the recorder ahead of the payload, drained first-in-first-out at the payload rate (mission volume 1500 GB/day (eoPortal Sentinel-2: 1.5 TB/day after compression, 2.4 Tbit memory)). Transmitter duty cycle: 30 min per orbit on the payload link. The TT&C route uses the 43 S-band stations of the network, has no recorder queue and spends no payload-transmitter time.

Direct downlink to the station network: 65.87 ms





term	time	share	note
wait for a pass over the network	0.0 µs	0.0 %	next: Oregon
radio propagation (slant range)	6.21 ms	9.4 %	1862 km at closest approach
recorder backlog drained ahead of the payload (FIFO)	0.0 µs	0.0 %	nothing ahead of it (recorder empty or priority)
serialisation at 520 Mbit/s	17.2 µs	0.0 %	one CCSDS frame minimum (1,115 B)
fibre Oregon -> Glasgow (Strathclyde)	59.64 ms	90.5 %	7183 km great circle × 1.5
ground processing / forwarding	0.0 µs	0.0 %	

Total 65.87 ms; the payload arrives at 2026-11-06 20:29:23 UTC. Space-segment rate 520.0 Mbit/s.

GEO relay (EDRS): 35 min 52 s





term	time	share	note
wait for line of sight to EDRS-C	35 min 52 s	100.0 %	
LEO -> GEO propagation	126.04 ms	0.0 %	~37786 km
GEO -> gateway propagation	130.71 ms	0.0 %	39187 km to Harwell (EDRS gateway)
serialisation at 1800 Mbit/s	5.0 µs	0.0 %	
fibre Harwell (EDRS gateway) -> Glasgow (Strathclyde)	9.84 ms	0.0 %	514 km
ground processing / forwarding	0.0 µs	0.0 %	

Total 35 min 52 s; the payload arrives at 2026-11-06 21:05:15 UTC. Space-segment rate 1.80 Gbit/s.

Own antenna at the data centre (hypothetical): 1 h 16 min



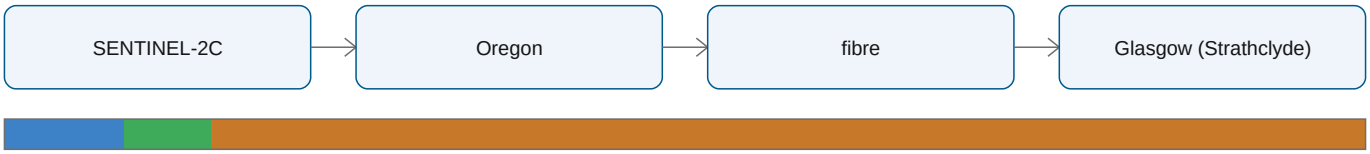


term	time	share	note
wait for a pass over Glasgow (Strathclyde)	1 h 16 min	100.0 %	
radio propagation	2.67 ms	0.0 %	801 km

term	time	share	note
recorder backlog drained ahead of the payload (FIFO)	0.0 µs	0.0 %	nothing ahead of it
serialisation at 520 Mbit/s	17.2 µs	0.0 %	
ground processing / forwarding	0.0 µs	0.0 %	

Total 1 h 16 min; the payload arrives at 2026-11-06 21:45:17 UTC. Space-segment rate 520.0 Mbit/s.

S-band TT&C channel (semantic message in telemetry): 70.31 ms



term	time	share	note
wait for a pass over any S-band station (43)	0.0 µs	0.0 %	next: Oregon
radio propagation	6.21 ms	8.8 %	1862 km
serialisation at 2.0 Mbit/s (telemetry channel)	4.46 ms	6.3 %	no recorder queue, no X-band transmitter time
fibre Oregon -> Glasgow (Strathclyde)	59.64 ms	84.8 %	7183 km
ground processing / forwarding	0.0 µs	0.0 %	

Total 70.31 ms; the payload arrives at 2026-11-06 20:29:23 UTC. Space-segment rate 2.0 Mbit/s.

LEO short-message relay (Iridium SBD-class modem, hypothetical fit): 5.22 s



term	time	share	note
wait: none (LEO relay constellation always overhead)	0.0 µs	0.0 %	Iridium: 66 satellites, global
message delivery through the relay (1 message of <= 1,960 B)	5.15 s	98.7 %	measured SBD latency 5-20 s per message
fibre Iridium gateway (Tempe, AZ) -> Glasgow (Strathclyde)	65.56 ms	1.3 %	7976 km
ground processing / forwarding	0.0 µs	0.0 %	

Total 5.22 s; the payload arrives at 2026-11-06 20:29:28 UTC. Space-segment rate 784 bit/s.

The payload on the radio

quantity	value
Information bits	159 bit
Channel bits actually sent	8.9 kbit (CCSDS AOS frames of 1115 bytes; one frame minimum)
Serialisation at 520.0 Mbit/s	0.3 µs for the payload bits; 17.2 µs framed
P(at least one bit error), coded link	1.59e-10
P(at least one bit error), uncoded QPSK	0.00e+0
Same payload on the S-band TT&C channel (2.0 Mbit/s)	4.46 ms

Reading: for a semantic message of a few hundred bits the radio time is microseconds and the fibre tens of milliseconds; the delay is decided by whether a station or a relay is in view when the message is created. The frog link gains nothing from a faster radio and everything from an always-on low-rate channel.

3. Link budget and noise

Computed for the link now, to Fairbanks (ASF) (NASA NEN).

quantity	value	how
Frequency	8.0900 GHz	catalogue

quantity	value	how
Slant range / elevation	1465.8 km / 27.60 deg	SGP4 position vs station
One-way propagation delay	4.889 ms	range / c
Doppler shift	-116.13 kHz	range-rate 4.303 km/s x f / c
Satellite EIRP	24.0 dBW (assumed)	catalogue
Free-space path loss	-173.93 dB	32.45 + 20 log f[MHz] + 20 log d[km]
Atmospheric gases	-0.11 dB	ITU-R P.676 zenith value / sin(el)
Rain 0 mm/h	-0.00 dB	ITU-R P.838 k R^alpha x effective path (3 km rain height)
Pointing + polarisation	-0.8 dB	0.5 + 0.3 dB assumed
Ground antenna 11 m	+57.17 dBi	10 log(eta (pi D / lambda)^2), eta 0.6
G/T (Tsys 150 K)	35.41 dB/K	G - 10 log Tsys
Received carrier power	-93.66 dBW (430.199 pW)	EIRP + G - losses
Noise density N0 = k Tsys	-206.84 dBW/Hz	k = 1.380649e-23 J/K
Occupied bandwidth	351.00 MHz	260.00 Msym/s x (1 + 0.35 roll-off)
Noise power in band	-121.39 dBW	N0 + 10 log B
C/N0	113.18 dB-Hz	EIRP + G/T - losses + 228.6
SNR in band	27.72 dB	C - N
Eb/N0 at 520.0 Mbit/s	26.02 dB	C/N0 - 10 log Rb
Required Eb/N0 + implementation	8.00 dB	8PSK, r=0.67 (CCSDS 130.1-G) + 1.5 dB
LINK MARGIN	18.02 dB (OK)	
BER uncoded QPSK	4.22e-176	0.5 erfc(sqrt(Eb/N0))
BER coded (model)	1.00e-12	1e-6 at the requirement, one decade per 0.3 dB
Shannon capacity in this band	3.23 Gbit/s	B log2(1 + SNR)
Nominal rate / capacity	16.1 %	

4. Contacts in the next 48 h

Stations in view now (EL >= 5 deg): Fairbanks (ASF) EL 27.6 deg, 1466 km; Inuvik EL 21.9 deg, 1681 km; Oregon EL 16.7 deg, 1931 km; Prince Albert EL 11.2 deg, 2276 km. GEO relay: not in line of sight.

AOS (UTC)	LOS	station	network	dur	max EL	min range	AZ AOS->LOS
11-06 20:29:15	20:31:42	Prince Albert	CSA / NRCan	2.5 min	11 deg	2256 km	292 -> 270
11-06 20:29:15	20:32:13	Inuvik	CSA / DLR / SSC	3.0 min	23 deg	1631 km	189 -> 195
11-06 20:29:15	20:33:15	Fairbanks (ASF)	NASA NEN	4.0 min	29 deg	1431 km	142 -> 171
11-06 20:29:15	20:35:25	Oregon	AWS Ground Station	6.2 min	18 deg	1862 km	320 -> 250
11-06 20:33:49	20:45:31	Hawaii (Oahu)	AWS Ground Station / NOAA	11.7 min	35 deg	1257 km	30 -> 170
11-06 21:01:13	21:13:59	McMurdo	NASA / NOAA	12.8 min	76 deg	840 km	33 -> 218
11-06 21:10:07	21:20:51	TrollSat (Antarctica)	KSAT	10.7 min	21 deg	1745 km	160 -> 44
11-06 21:19:56	21:32:28	Cape Town	AWS Ground Station	12.5 min	67 deg	864 km	163 -> 354
11-06 21:21:50	21:33:45	Hartebeesthoek	SANSA	11.9 min	40 deg	1153 km	181 -> 326
11-06 21:40:58	21:52:20	Matera	e-GEOS / ASI (Copernicus core)	11.4 min	29 deg	1420 km	198 -> 330
11-06 21:41:43	21:54:13	Toulouse (Aussaguel)	CNES	12.5 min	84 deg	798 km	162 -> 347
11-06 21:41:52	21:48:38	Maspalomas	ESA / INTA	6.8 min	10 deg	2377 km	99 -> 32
11-06 21:44:26	21:56:10	Neustrelitz	DLR	11.7 min	33 deg	1300 km	195 -> 335
11-06 21:44:46	21:57:13	Ireland	AWS Ground Station	12.4 min	68 deg	854 km	152 -> 347
11-06 21:46:30	21:57:28	Stockholm	AWS Ground Station	11.0 min	24 deg	1597 km	210 -> 332
11-06 21:49:06	21:59:52	Kiruna (ESTRACK)	ESA	10.8 min	22 deg	1673 km	217 -> 336
11-06 21:49:07	21:59:52	Esrange (Kiruna)	SSC	10.7 min	22 deg	1677 km	217 -> 336
11-06 21:49:27	22:00:31	Tromsø	KSAT	11.1 min	24 deg	1580 km	213 -> 337

AOS (UTC)	LOS	station	network	dur	max EL	min range	AZ AOS->LOS
11-06 21:51:39	22:03:12	Svalbard (SvalSat)	KSAT	11.5 min	29 deg	1412 km	208 -> 342
11-06 21:59:17	22:11:25	Inuvik	CSA / DLR / SSC	12.1 min	47 deg	1045 km	27 -> 237
11-06 22:00:48	22:13:14	Fairbanks (ASF)	NASA NEN	12.4 min	64 deg	878 km	20 -> 218
11-06 22:13:58	22:23:36	Hawaii (Oahu)	AWS Ground Station / NOAA	9.6 min	18 deg	1876 km	340 -> 238
11-06 22:33:49	22:40:17	Dubbo	AWS Ground Station	6.5 min	9 deg	2441 km	86 -> 149
11-06 22:41:32	22:53:54	McMurdo	NASA / NOAA	12.4 min	43 deg	1121 km	357 -> 205
11-06 22:49:55	23:02:24	TrollSat (Antarctica)	KSAT	12.5 min	49 deg	1031 km	156 -> 359
11-06 23:00:50	23:09:15	Cape Town	AWS Ground Station	8.4 min	13 deg	2159 km	201 -> 286
11-06 23:17:49	23:30:16	Maspalomas	ESA / INTA	12.4 min	71 deg	831 km	172 -> 344
11-06 23:23:58	23:32:10	Toulouse (Aussaguel)	CNES	8.2 min	12 deg	2206 km	230 -> 314
11-06 23:25:33	23:36:20	Ireland	AWS Ground Station	10.8 min	23 deg	1642 km	210 -> 330
11-06 23:33:15	23:39:40	Tromsø	KSAT	6.4 min	9 deg	2468 km	269 -> 331
11-06 23:33:37	23:43:30	Svalbard (SvalSat)	KSAT	9.9 min	17 deg	1911 km	247 -> 352
11-06 23:33:51	23:38:20	Kiruna (ESTRACK)	ESA	4.5 min	7 deg	2639 km	280 -> 323
11-06 23:33:53	23:38:19	Esrange (Kiruna)	SSC	4.4 min	7 deg	2642 km	280 -> 323
11-06 23:38:33	23:49:28	Inuvik	CSA / DLR / SSC	10.9 min	26 deg	1543 km	36 -> 275
11-06 23:40:23	23:51:17	Fairbanks (ASF)	NASA NEN	10.9 min	26 deg	1539 km	22 -> 261
11-06 23:54:14	23:59:58	Hatoyama (EOC)	JAXA	5.7 min	8 deg	2512 km	61 -> 117
11-07 00:09:26	00:22:02	Dubbo	AWS Ground Station	12.6 min	85 deg	803 km	13 -> 193
11-07 00:09:40	00:19:22	Alice Springs	Geoscience Australia	9.7 min	18 deg	1887 km	59 -> 161
11-07 00:22:50	00:34:03	McMurdo	NASA / NOAA	11.2 min	24 deg	1612 km	319 -> 195
11-07 00:29:38	00:42:20	TrollSat (Antarctica)	KSAT	12.7 min	75 deg	841 km	150 -> 319
11-07 00:51:49	00:55:36	Cuiabá	INPE	3.8 min	6 deg	2662 km	101 -> 65
11-07 01:08:00	01:16:13	Gatineau	NRCan	8.2 min	13 deg	2169 km	97 -> 14
11-07 01:08:03	01:12:30	Wallops	NASA NEN	4.5 min	7 deg	2612 km	78 -> 35
11-07 01:16:28	01:24:08	Svalbard (SvalSat)	KSAT	7.7 min	11 deg	2317 km	288 -> 4
11-07 01:17:07	01:27:20	Inuvik	CSA / DLR / SSC	10.2 min	21 deg	1749 km	54 -> 305
11-07 01:19:25	01:28:24	Fairbanks (ASF)	NASA NEN	9.0 min	15 deg	2033 km	32 -> 299
11-07 01:30:54	01:43:24	Hatoyama (EOC)	JAXA	12.5 min	88 deg	794 km	13 -> 194
11-07 01:31:33	01:42:53	Seoul	AWS Ground Station	11.3 min	28 deg	1431 km	31 -> 162
11-07 01:32:36	01:40:22	Miyun	CRESDA / RADI	7.8 min	11 deg	2271 km	50 -> 128
11-07 01:47:41	01:59:33	Alice Springs	Geoscience Australia	11.9 min	37 deg	1214 km	351 -> 208
11-07 01:50:07	02:02:02	Dongara	SSC	11.9 min	40 deg	1167 km	36 -> 179
11-07 01:53:35	01:57:52	Dubbo	AWS Ground Station	4.3 min	7 deg	2646 km	290 -> 250
11-07 02:05:13	02:14:26	McMurdo	NASA / NOAA	9.2 min	14 deg	2109 km	278 -> 184
11-07 02:09:03	02:21:09	TrollSat (Antarctica)	KSAT	12.1 min	41 deg	1157 km	139 -> 282
11-07 02:18:06	02:28:13	Punta Arenas	SSC / KSAT / AWS	10.1 min	18 deg	1879 km	146 -> 39
11-07 02:23:21	02:35:08	Córdoba	CONAE	11.8 min	35 deg	1274 km	152 -> 12
11-07 02:26:58	02:39:25	Cuiabá	INPE	12.5 min	71 deg	838 km	173 -> 341
11-07 02:42:31	02:54:52	Wallops	NASA NEN	12.4 min	63 deg	879 km	155 -> 352
11-07 02:43:59	02:55:41	Ohio	AWS Ground Station	11.7 min	37 deg	1220 km	139 -> 359
11-07 02:44:29	02:56:59	Gatineau	NRCan	12.5 min	78 deg	811 km	160 -> 348
11-07 02:47:21	02:56:27	Sioux Falls (EROS)	USGS Landsat	9.1 min	16 deg	1995 km	106 -> 11
11-07 02:50:21	02:59:43	Prince Albert	CSA / NRCan	9.4 min	16 deg	1947 km	99 -> 1
11-07 02:55:00	03:05:52	Inuvik	CSA / DLR / SSC	10.9 min	25 deg	1559 km	84 -> 324
11-07 02:57:17	03:06:02	Fairbanks (ASF)	NASA NEN	8.8 min	14 deg	2076 km	55 -> 325
11-07 02:59:28	03:05:29	Svalbard (SvalSat)	KSAT	6.0 min	8 deg	2514 km	324 -> 22
11-07 03:10:17	03:22:47	Miyun	CRESDA / RADI	12.5 min	82 deg	801 km	12 -> 198

AOS (UTC)	LOS	station	network	dur	max EL	min range	AZ AOS->LOS
11-07 03:10:55	03:22:19	Seoul	AWS Ground Station	11.4 min	31 deg	1351 km	357 -> 225
11-07 03:12:20	03:19:18	Hatoyama (EOC)	JAXA	7.0 min	10 deg	2354 km	334 -> 265
11-07 03:16:38	03:29:07	Sanya	CRESDA / RADI	12.5 min	87 deg	792 km	13 -> 192
11-07 03:21:33	03:33:59	Singapore	KSAT / AWS	12.4 min	72 deg	827 km	17 -> 188
11-07 03:30:38	03:40:33	Dongara	SSC	9.9 min	18 deg	1872 km	328 -> 222
11-07 03:48:00	03:59:35	TrollSat (Antarctica)	KSAT	11.6 min	32 deg	1377 km	121 -> 252
11-07 03:48:21	03:55:14	McMurdo	NASA / NOAA	6.9 min	9 deg	2462 km	238 -> 171
11-07 03:56:58	04:09:39	Punta Arenas	SSC / KSAT / AWS	12.7 min	84 deg	814 km	165 -> 340
11-07 04:03:09	04:13:44	Córdoba	CONAE	10.6 min	23 deg	1660 km	191 -> 306
11-07 04:23:46	04:34:55	Ohio	AWS Ground Station	11.2 min	26 deg	1500 km	201 -> 329
11-07 04:24:07	04:32:57	Wallops	NASA NEN	8.8 min	14 deg	2087 km	223 -> 315
11-07 04:24:28	04:36:58	Sioux Falls (EROS)	USGS Landsat	12.5 min	78 deg	810 km	169 -> 345
11-07 04:26:17	04:35:18	Gatineau	NRCan	9.0 min	14 deg	2062 km	225 -> 318
11-07 04:27:28	04:39:56	Prince Albert	CSA / NRCan	12.5 min	72 deg	832 km	155 -> 347

Passes over the data centre itself (Glasgow (Strathclyde)): 11-06 21:45 (84 deg, 13 min); 11-06 23:26 (20 deg, 10 min); 11-07 09:50 (18 deg, 10 min); 11-07 11:29 (85 deg, 13 min); 11-07 13:09 (23 deg, 11 min); 11-07 14:50 (7 deg, 4 min); 11-07 19:38 (15 deg, 9 min); 11-07 21:15 (53 deg, 12 min); 11-07 22:55 (32 deg, 12 min); 11-08 09:21 (11 deg, 8 min); 11-08 10:59 (54 deg, 12 min); 11-08 12:39 (33 deg, 12 min); 11-08 14:19 (10 deg, 7 min); 11-08 19:10 (10 deg, 7 min).

GEO relay line-of-sight windows: EDRS-C 11-06 21:05 - 22:05; EDRS-A 11-06 21:05 - 22:05; EDRS-C 11-06 22:43 - 23:46; EDRS-A 11-06 22:45 - 23:45; EDRS-C 11-07 00:18 - 04:09; EDRS-A 11-07 00:23 - 01:27; EDRS-A 11-07 01:58 - 05:48; EDRS-C 11-07 04:38 - 05:42; EDRS-A 11-07 06:19 - 07:22; EDRS-C 11-07 06:20 - 07:20; EDRS-C 11-07 08:00 - 09:00; EDRS-A 11-07 08:00 - 09:01.

5. Same routes, every payload

payload	bits	direct X-band	GEO relay	own antenna	S-band TT&C	SBD relay
Frog Class-4 alarm message (event flag + region)	64 bit	65.87 ms	35 min 52 s	1 h 16 min	70.31 ms	5.13 s
Frog fibres, one frame (EO sim)	159 bit	65.87 ms	35 min 52 s	1 h 16 min	70.31 ms	5.22 s
Frog fibres, one frame (CDnet @25 fps)	321 bit	65.87 ms	35 min 52 s	1 h 16 min	70.31 ms	5.38 s
Frog fibres, 5-min observation (300 frames)	48.0 kbit	65.95 ms	35 min 52 s	1 h 16 min	89.85 ms	1 min 20 s
Frog fibres + evidence crops, 5 min	109.0 kbit	66.06 ms	35 min 52 s	1 h 16 min	120.35 ms	2 min 20 s
JPEG q50 frame 320×240	23.4 kbit	65.90 ms	35 min 52 s	1 h 16 min	77.56 ms	40.07 s
H.264 frame, 8 kbit/frame setting	8.2 kbit	65.87 ms	35 min 52 s	1 h 16 min	70.31 ms	12.92 s
JPEG q50, 5-min observation	7.03 Mbit	79.36 ms	35 min 52 s	1 h 16 min	3.58 s	n/a
Raw 8-bit frame 320×240	614.4 kbit	67.03 ms	35 min 52 s	1 h 16 min	373.05 ms	13 min 20 s
Sentinel-2 L1C tile (100 km², ~600 MB)	4.80 Gbit	9.30 s	35 min 55 s	1 h 16 min	55 min 42 s (8 p)	n/a
Landsat 8/9 L1 scene (~1 GB)	8.00 Gbit	15.45 s	35 min 57 s	1 h 16 min	1 h 30 min (16 p)	n/a
WorldView-3 pan+MS strip, 13 × 100 km (~4 GB)	32.00 Gbit	1 min 2 s	36 min 10 s	1 h 17 min	5 h 43 min (49 p)	n/a
Sentinel-1 IW SLC product (~8 GB)	64.00 Gbit	2 min 3 s	36 min 28 s	1 h 18 min	11 h 24 min (94 p)	n/a

6. Methods, assumptions and sources

Orbits: CelesTrak GP element sets (US Space Force catalogue) propagated with SGP4 (satellite.js). Accuracy about 1 km at the element epoch, degrading with age. Positions and pass times were cross-checked against the independent Python sgp4 package (agreement < 0.06 deg, pass times within 10 s) and against Heavens-Above (same passes; it uses a 10 deg mask).

Contacts: 48 h scan at 20 s with AOS/LOS refined by bisection to under 1 s, above the chosen elevation mask. Target opportunities use the elevation equivalent to the maximum off-nadir angle: cos(el) = sin(eta) (R + h) / R. GEO relay line of sight: the LEO-GEO segment must clear the Earth by 60 km.

Delivery time = waiting for the next contact (multi-pass drain for large files, at the nominal rate) + propagation (range / c) + serialisation (payload bits in CCSDS AOS frames of 1,115 bytes, one frame minimum, at the nominal rate) + terrestrial fibre (great-circle distance x 1.5 route factor at 2/3 c plus 0.5 ms per hop for 12 hops) + the ground processing time entered. GEO relay: LEO-GEO about 37,800 km, GEO-gateway from geometry, relay rate from the catalogue, fibre from the gateway.

Link budget: free-space loss, ITU-R P.676 gaseous absorption (zenith value / sin el), ITU-R P.838 rain (k, alpha at the band; 3 km rain height; P.618 reduction), pointing 0.5 dB, polarisation 0.3 dB, implementation 1.5 dB, antenna efficiency 0.6, roll-off 0.35, required Eb/N0 from CCSDS 130.1-G performance curves. Uncoded BER = 0.5 erfc(sqrt(Eb/N0)); coded BER is a waterfall model (1e-6 at the requirement, one decade per 0.3 dB, floor 1e-12), indicative only.

Why real missions quote hours (docs/LATENCY_RESEARCH.md): a satellite has about 10 minutes of ground contact per 100-minute orbit and 4-6 windows a day with any one station; it collects 0.5-1.5 TB/day into its recorder, so imagery queues for hours behind earlier data and the X-band transmitter is duty-cycle limited (Sentinel-1: 30 min per orbit); then the product is processed under a timeliness policy (Sentinel NRT 1-3 h, standard 24 h; Landsat ~5 h). The simulator models the queue (backlog, FIFO or priority), the duty cap and the processing time; a semantic message avoids the queue and the payload transmitter entirely on the TT&C route.

What is NOT modelled: station scheduling and contention, encryption and product processing, data policy (e.g. ESA's near-real-time commitment for Sentinel products is under 3 h, set by processing and policy rather than the radio), antenna slew and acquisition time (typically 1 to 2 minutes at the start of a pass), and the actual availability of relay service.

Confidence tags in the catalogue: SOURCED = operator or engineering source (listed on the card and in section 1); TYPICAL = secondary source (eoPortal, trade press), treat as +/-20 %; ASSUMED = class default. Satellite EIRP and the ground G/T are never published per mission and are always assumptions, so the margin and BER are indicative rather than a certified budget.

Payload sizes: the frog link, JPEG and H.264 figures are measured in this project's simulations (eo_satellite_sim, E8 held-out run); full product sizes from the operators (Copernicus, USGS, Maxar).

Stations used in this report

station	network	lat	lon	bands	dish	source
Svalbard (SvalSat)	KSAT	78.23	15.39	S/X/Ka	13 m	KSAT: 78°N, 160+ antennas, up to 14 passes/day for polar orbits
Tromsø	KSAT	69.66	18.94	S/X/Ka	11 m	KSAT Tromsø
TrollSat (Antarctica)	KSAT	-72.01	2.54	S/X/Ka	7.3 m	KSAT TrollSat 72°S
Inuvik	CSA / DLR / SSC	68.32	-133.55	S/X	13 m	Inuvik Satellite Station Facility
Kiruna (ESTRACK)	ESA	67.86	20.97	S/X	15 m	ESA: KIR-2 67.858428°N 20.966880°E, S/X-band
Esrange (Kiruna)	SSC	67.88	21.07	S/X/Ka	13 m	SSC Esrange
Fairbanks (ASF)	NASA NEN	64.86	-147.85	S/X	11 m	Alaska Satellite Facility
McMurdo	NASA / NOAA	-77.85	166.67	S/X/Ka	10 m	NASA McMurdo Ground Station
Neustrelitz	DLR	53.33	13.07	S/X	7.3 m	DLR Neustrelitz
Matera	e-GEOS / ASI (Copernicus core)	40.65	16.70	X	10 m	Matera 40°382573N 16°422153E, Sentinel X-band station
Maspalomas	ESA / INTA	27.76	-15.63	S/X	15 m	ESA: 15 m antenna 27°452463N 15°382023W
Sioux Falls (EROS)	USGS Landsat	43.73	-96.62	S/X	10 m	USGS EROS Landsat Ground Station
Wallops	NASA NEN	37.94	-75.47	S/X	11 m	NASA Wallops
Hartebeesthoek	SANSA	-25.89	27.69	S/X	12 m	SANSA Hartebeesthoek
Punta Arenas	SSC / KSAT / AWS	-53.00	-70.85	S/X	7.3 m	SSC Punta Arenas; AWS Ground Station
Alice Springs	Geoscience Australia	-23.76	133.88	S/X	9 m	GA Alice Springs (Landsat / Sentinel)
Dongara	SSC	-29.05	115.35	S/X	7.3 m	SSC Western Australia Space Centre
Dubbo	AWS Ground Station	-32.24	148.60	S/X	5.4 m	AWS Ground Station Dubbo
Singapore	KSAT / AWS	1.35	103.82	S/X	7.3 m	KSAT Singapore; AWS Ground Station
Dubai	KSAT	25.20	55.30	S/X	7.3 m	KSAT Dubai
Bahrain	AWS Ground Station	26.00	50.60	S/X	5.4 m	AWS Ground Station Bahrain
Mauritius	KSAT	-20.30	57.60	S/X	7.3 m	KSAT Mauritius
Hawaii (Oahu)	AWS Ground Station / NOAA	21.30	-158.00	S/X	5.4 m	AWS Ground Station Hawaii
Oregon	AWS Ground Station	45.90	-119.30	S/X	5.4 m	AWS Ground Station Oregon
Ohio	AWS Ground Station	40.40	-82.90	S/X	5.4 m	AWS Ground Station Ohio

station	network	lat	lon	bands	dish	source
Ireland	AWS Ground Station	53.30	-6.30	S/X	5.4 m	AWS Ground Station Ireland
Stockholm	AWS Ground Station	59.30	18.10	S/X	5.4 m	AWS Ground Station Stockholm
Seoul	AWS Ground Station	37.50	127.00	S/X	5.4 m	AWS Ground Station Seoul
Cape Town	AWS Ground Station	-33.90	18.40	S/X	5.4 m	AWS Ground Station Cape Town
Prince Albert	CSA / NRCan	53.21	-105.93	S/X	10 m	Prince Albert Satellite Station
Gatineau	NRCan	45.58	-75.81	S/X	10 m	Gatineau Satellite Station
Shadnagar (NRSC)	ISRO	17.03	78.18	S/X	10 m	ISRO NRSC Shadnagar
Hatoyama (EOC)	JAXA	36.00	139.35	S/X	10 m	JAXA Earth Observation Center
Miyun	CRESDA / RADI	40.40	116.90	S/X	12 m	RADI Miyun station
Kashi	CRESDA / RADI	39.50	76.00	S/X	12 m	RADI Kashi station
Sanya	CRESDA / RADI	18.30	109.30	S/X	12 m	RADI Sanya station
Cuiabá	INPE	-15.55	-56.07	S/X	11 m	INPE Cuiabá
Córdoba	CONAE	-31.52	-64.46	S/X	7.3 m	CONAE Córdoba
Toulouse (Aussaguel)	CNES	43.43	1.50	S/X	10 m	CNES Aussaguel

Verification links: https://www.n2yo.com/satellite/?s=60989 ;
https://www.heavens-above.com/PassSummary.aspx?satid=60989&lat=55.8600&lng=-4.2400&alt=0&tz=UCT&showAll=t ;
https://celestrak.org/NORAD/elements/gp.php?CATNR=60989&FORMAT=tle